Jerby-Arnon 2021 synovial sarcoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO (scoped to SmartSeq2)

Two platforms (SmartSeq2 TPM, 12 pt; 10x UMI, 3 pt) are an irreducible batch — technology iLISI stays 1.0/2 under every Harmony strength and batch key. The SmartSeq2 compartment is one continuous, patient-integrated manifold (iLISI 1.0→6.2 of 11 patients, single component); 10x has only 3 patients and fragments. Per the biology-before-geometry rule the trajectory is scoped to SmartSeq2.

Gate 2 — module geography → GO (non-branching)

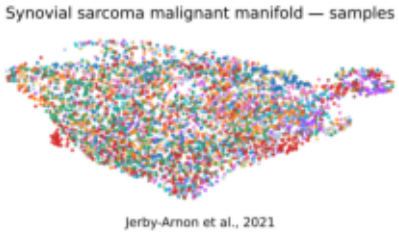
Signatures collapse onto one axis: smoothed Gast & pSom co-vary (r=+0.23) and both oppose Crest (−0.37 / −0.35). Along pseudotime Gast declines (ρ −0.47) and Crest rises to a terminal pole (ρ +0.48; Crest-high decile at t=0.71). No Gast-vs-pSom split → non-branching (still a GO). Crest enriches the terminus, not the Gast origin → no inversion flag.

Gate 3 — cross-repression → NA

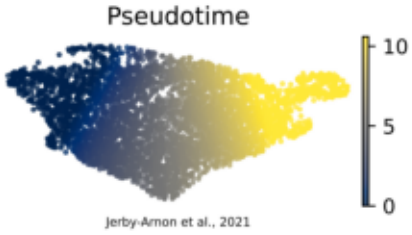
No bifurcation: a PPT tree only wallpapers the compact cloud (forks grow 6→9→15 with node count), while the fold-free principal curve has 0 forks. With no two arms there is no hub to test for mutual Gast↔pSom repression.

STEP FIGURES

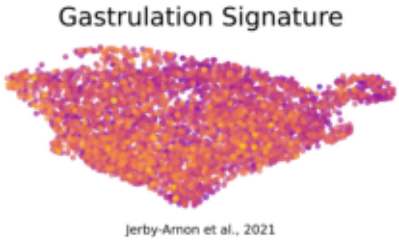
Manifold — samples



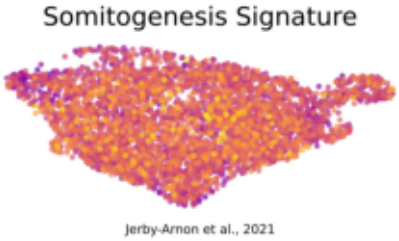
Pseudotime



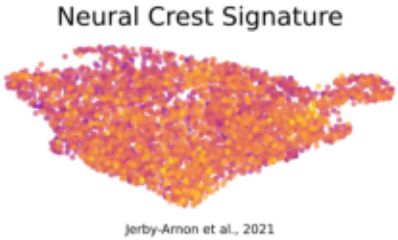
Gastrulation



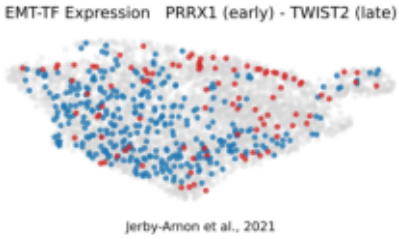
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Synovial sarcoma (Jerby-Arnon 2021); SmartSeq2 + 10x. 12,694 malignant cells merged; trajectory on the 4,338-cell SmartSeq2 manifold (11 patients after thin-sample removal).

Verdict.
The embryonic programme reproduces as a single non-branching developmental axis — a Gast/pSom-high origin flowing to a neural-crest-high terminus, with cell cycle orthogonal (ρ −0.01). This is the Linear-pSom/Crest geometry (pSom globally dominant, mean +0.32). No hub, no cross-repressing arms → Gate 3 NA. EMT axis: PRRX1 (early/root) → TWIST2 (late/terminus).

Parameter choices

Integration scope	SmartSeq2 (technology irreducible: iLISI_tech=1.0/2)	**Sample threshold**	≥98 cells (mult-gap 296→33, 9.0×); drop SyS10
Malignant cells	12,694 merged → 4,338 SmartSeq2 (11 patients)	**Integration**	Harmony θ=10 on sample (iLISI 1.0→6.2/11); PCA20, spread 1.3
Geometry	Linear-pSom/Crest (0 forks, fold ratio 2.95)	**Root**	curve tip 0 = Gast/pSom-high, Crest-low
Gene recovery	Gast 78/81 · pSom 95/98 · Crest 102/104	**EMT-TFs**	PRRX1 (root) → TWIST2 (terminus)